

GARI, Russian Federation

WMO: 280490

Lat: 59.433N

Lon: 62.333E

Elev: 131

StdP: 99.76

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.8	-30.2	-35.6	0.1	-32.6	-33.0	0.2	-30.1	10.1	-13.2	9.2	-12.1	3.0	0	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	28.9	19.7	27.2	18.8	25.3	17.8	21.0	27.0	19.9	25.7	18.8	23.7	3.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.9	24.0	17.7	12.9	22.6	16.6	12.0	21.7	61.3	27.0	57.5	25.8	53.8	23.8	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.5	7.6	-35.0	31.7	3.9	1.8	-37.8	33.0	-40.1	34.1	-42.3	35.2	-45.1	36.5		
			-35.1	22.7	3.9	1.5	-37.9	23.8	-40.2	24.7	-42.3	25.5	-45.2	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-15.9	-13.7	-4.9	2.8	10.4	15.9	18.2	15.0	9.2	2.1	-7.9	-13.4
	DBStd	13.29	7.38	7.75	5.28	5.31	5.49	4.80	4.49	4.27	4.41	5.13	7.61	8.08
	HDD10.0	3808	804	663	461	221	64	8	2	5	67	250	538	727
	HDD18.3	6231	1062	896	719	466	251	103	60	122	276	504	788	985
	CDD10.0	727	0	0	0	5	77	184	254	159	42	4	0	0
	CDD18.3	109	0	0	0	0	6	29	55	18	1	0	0	0
	CDH23.3	858	0	0	0	0	56	243	426	127	6	0	0	0
	CDH26.7	197	0	0	0	0	5	56	102	35	0	0	0	0
Wind	WSAvg	3.8	3.9	3.9	4.3	4.1	4.2	3.9	3.3	3.4	3.6	3.8	3.8	3.8
Precipitation	PrecAvg	510	23	14	20	23	45	68	76	84	72	36	27	22
	PrecMax	761	55	36	38	46	76	168	184	188	162	71	50	41
	PrecMin	357	2	2	1	0	17	10	3	21	19	10	6	7
	PrecStd	76	11	8	11	11	18	40	37	40	37	19	13	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	2.5	8.0	19.6	27.4	30.5	31.2	30.4	24.1	16.5	6.5	2.2
		MCWB	0.6	0.3	3.2	10.9	15.2	19.9	20.9	21.1	16.0	11.0	5.0	0.9
	2%	DB	-1.7	0.7	5.4	15.7	25.0	28.2	29.2	26.6	20.8	14.0	4.3	0.6
		MCWB	-2.6	-0.8	1.7	9.1	14.5	18.6	20.3	19.4	14.7	9.7	3.1	-0.6
	5%	DB	-3.8	-1.2	3.6	13.1	22.3	26.2	27.6	23.9	18.1	11.7	2.5	-1.3
		MCWB	-4.4	-2.4	0.5	7.3	13.1	17.9	19.6	18.0	13.3	8.4	1.2	-2.2
	10%	DB	-6.0	-3.3	2.0	10.8	19.6	24.0	25.8	21.8	15.6	9.4	0.8	-3.2
		MCWB	-6.6	-4.6	-0.3	5.6	11.9	16.8	18.9	16.8	12.2	6.9	-0.3	-4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.3	0.8	3.9	11.8	16.9	21.5	22.6	21.9	17.5	11.7	5.1	1.1
		MCDB	1.1	2.0	6.8	17.4	24.0	27.8	27.8	27.8	22.1	15.3	6.2	2.0
	2%	WB	-2.5	-0.6	2.2	9.7	15.1	19.9	21.4	20.1	15.2	10.2	3.2	-0.5
		MCDB	-1.8	0.7	4.6	15.1	23.0	26.3	27.1	25.2	19.6	13.2	4.2	0.5
	5%	WB	-4.4	-2.3	0.9	7.5	13.9	18.6	20.4	18.6	13.8	8.7	1.4	-2.1
		MCDB	-3.8	-1.2	3.2	12.3	20.8	24.4	26.3	23.0	17.3	11.5	2.3	-1.3
	10%	WB	-6.6	-4.4	-0.2	6.1	12.5	17.5	19.5	17.4	12.5	7.0	-0.2	-4.0
		MCDB	-6.0	-3.4	1.7	10.1	18.6	23.1	24.9	21.0	15.2	9.2	0.6	-3.3
Mean Daily Temperature Range	5% DB	MDBR	6.1	7.2	7.7	8.3	9.6	9.1	8.9	8.0	7.2	5.7	5.4	5.8
		MCDBR	7.0	6.5	8.5	11.4	13.0	11.7	10.7	10.9	10.4	8.3	5.5	6.0
		MCWBR	6.7	5.5	6.1	7.2	6.5	5.5	4.7	5.3	5.5	5.2	4.7	5.6
	5% WB	MCDDBR	7.1	6.5	7.3	10.8	11.8	10.8	10.4	9.7	9.3	7.8	5.1	6.0
		MCWBR	6.9	5.6	5.6	7.2	6.6	5.7	5.3	5.3	5.4	5.2	4.5	5.7
Clear-Sky Solar Irradiance	taub	0.266	0.302	0.355	0.393	0.393	0.394	0.408	0.418	0.367	0.331	0.285	0.256	
	taud	2.018	2.043	1.985	2.100	2.295	2.350	2.308	2.250	2.358	2.315	2.126	1.967	
	Ebn at Noon	493	678	748	797	835	839	814	770	755	658	474	352	
	Edh at Noon	50	85	123	132	117	113	115	113	86	66	45	36	
All-Sky Solar Radiation	RadAvg	0.49	1.40	2.79	4.09	5.09	5.41	5.38	3.93	2.36	1.21	0.50	0.25	
	RadStd	0.04	0.12	0.22	0.40	0.40	0.43	0.41	0.47	0.40	0.25	0.17	0.07	0.03

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page